MSGED Program
Department of Geography and Environment
Jahangirnagar University

MSGED 530: Urban Development Planning and Management

Fall Semester 2024
Marks 10
Date of Submission: 14.03.2025 during class hour.

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Home District Town: Sirajganj Sadar Town
Considering the ecological models of urban landuse, elaborate your view on the model that best fits your home district town. Give examples on why you choose this model.

Option 01: Concentric zone model.
Option 02: Sector model.
Option 03: Multiple nuclei model.

Answer:

Urban ecological models, designed to elucidate the spatial organization of cities, offer valuable frameworks for understanding the complex interplay of land use and development. While the concentric zone model, sector model, and multiple nuclei model each provide unique perspectives, it is the **sector model** that most accurately reflects the evolving land use patterns observed in my home district town. This assertion is not merely theoretical; it is grounded in empirical observations derived from satellite imagery, building density analysis, and local knowledge, all of which point to a radial pattern of urban expansion, a hallmark of the sector model.

The sector model, proposed by Homer Hoyt, posits that urban growth occurs in wedge-shaped sectors radiating outward from the central business district (CBD) along transportation routes. This model diverges from the concentric zone model's rigid ring structure, recognizing that certain land uses, such as industrial or high-income residential areas, tend to expand along specific corridors. My direct observation of land use and land cover changes, facilitated by a comparative analysis of Sentinel-2 satellite imagery from 2017 to 2025 (10mx10m resolution) [Figure 1], strongly supports this radial expansion. The imagery revealed a distinct radiative growth pattern, with urban development extending outward along major roads and transportation lines, mirroring the sector model's core principle.

Furthermore, my independent creation of a detailed land use map [Figure 2], incorporating building data density from the ESRI portal and my personal understanding of the town, corroborated the sector model's applicability. This map meticulously delineated various land use categories, including commercial, residential (high, high-mid, mid, low-mid, low), small-scale industry, and the rural-urban fringe. The spatial distribution of these categories demonstrated a clear sectoral pattern. Commercial and small-scale industrial activities were concentrated closest to the CBD, forming the nucleus of the radiating sectors. High-residential areas occupied a small portion of this central zone, while lower-class residential areas extended outward, also exhibiting a radiative tendency. Middle-class residential areas, located further from the CBD, demonstrated a similar sectoral expansion, forming distinct wedges along transportation routes.

The observed density variations within these sectors further reinforced the sector model's relevance. While some lower-class residential areas exhibited higher building densities, reflecting older, more concentrated development, the rural-urban fringe also displayed pockets of high density. However, the Sentinel-2 imagery [Figure 1] comparison revealed that these fringe densities were relatively recent, indicative of new, rapidly expanding development along transportation corridors. This pattern aligns with the sector model's emphasis on transportation infrastructure as a driver of urban growth.

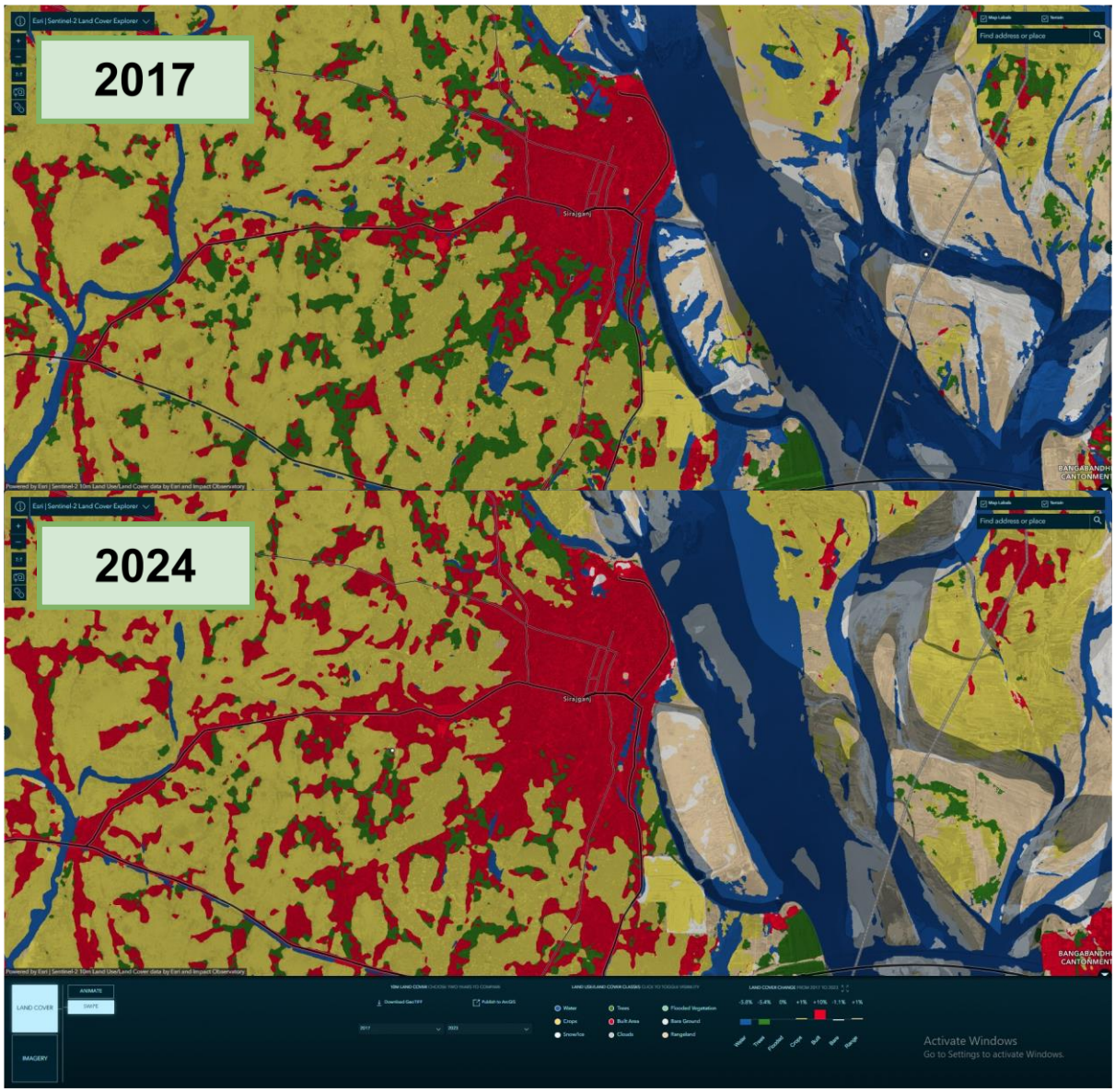

Figure 1: Land cover change from 2017 to 2024 (Sirajganj Sadar Town). Source: Sentinel-2 from livingatlas.arcgis.com

To validate my observations and analysis, I engaged in discussions with two long-time residents of the town. Their insights, grounded in years of lived experience, confirmed the sectoral trends I had identified. They described the gradual expansion of commercial and residential areas along specific roads, echoing the radial growth pattern evident in the satellite imagery and land use map.

In contrast, the concentric zone model, with its emphasis on uniform ring-like expansion, fails to capture the nuanced realities of my home district town. The multiple nuclei model, while acknowledging the existence of multiple centers of growth, does not adequately explain the dominant radial pattern observed. The sectoral growth, driven by transportation corridors and specific land use preferences, demonstrates that the sector model provides the most accurate and insightful framework for understanding the town's urban development.

In conclusion, the empirical evidence gathered from satellite imagery analysis, building density mapping, and local knowledge unequivocally supports the sector model as the most fitting representation of urban land use in my home district town. The radiating pattern of development, driven by transportation routes and specific land use preferences, underscores the model's ability to capture the dynamic and nuanced realities of urban growth. By embracing the sector model, we gain a deeper understanding of the forces shaping our urban landscape, enabling more informed and sustainable urban planning and development strategies

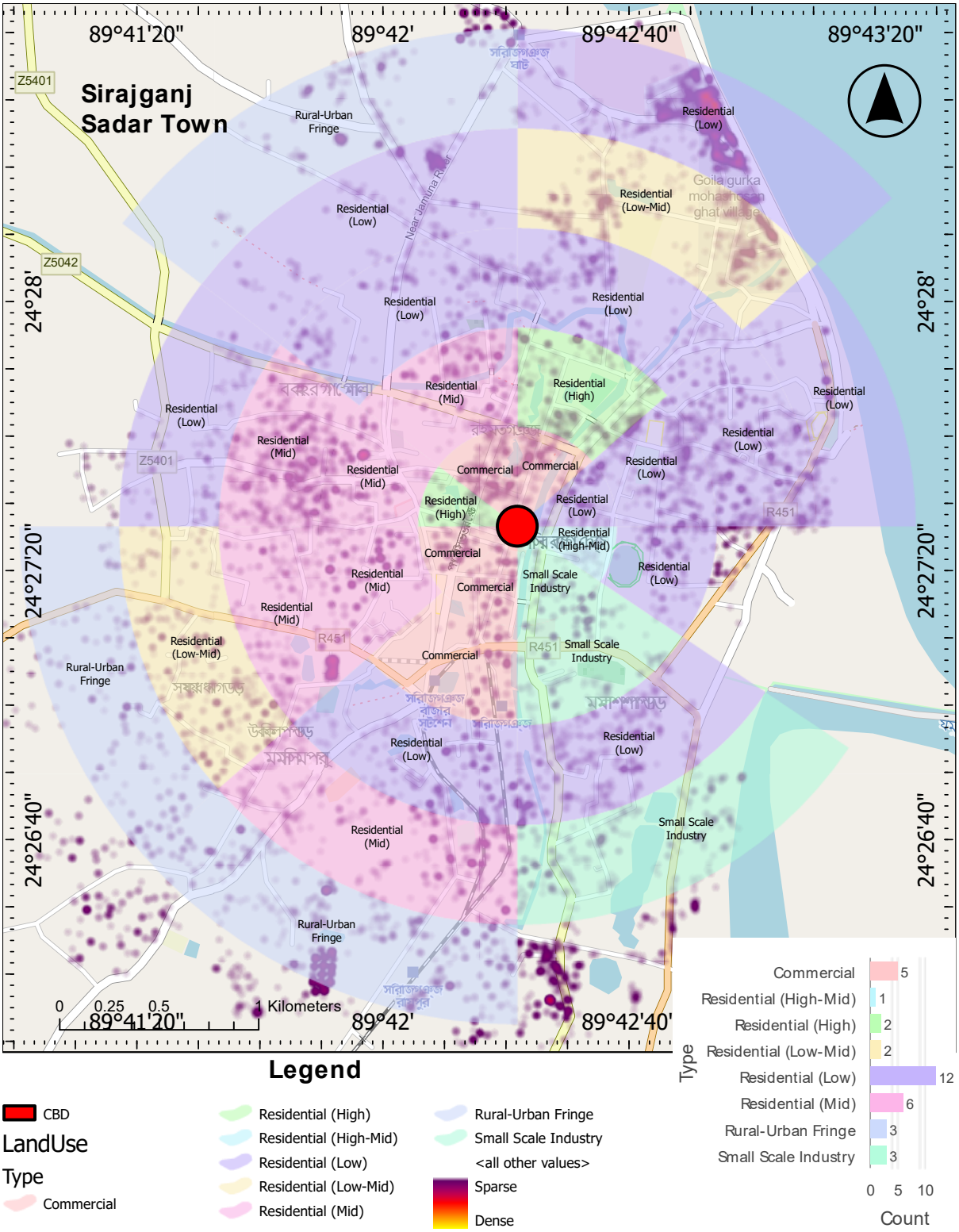

Figure 2: Land use map of Sirajganj Sadar Town. Sources: ESRI portal, Local opinion, Personal assumptions.